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from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

# Load dataset
iris = load_iris()
X = iris.data
y = iris.target

# Split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# Normalize
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

```

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# K-Nearest Neighbors (KNN)
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

# Model
knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train, y_train)

# Prediction
y_pred_knn = knn.predict(X_test)

# Accuracy
print("KNN Accuracy:", accuracy_score(y_test, y_pred_knn))

```

KNN Accuracy: 1.0

```

# Decision Tree

from sklearn.tree import DecisionTreeClassifier

# Model
dt = DecisionTreeClassifier(random_state=42)
dt.fit(X_train, y_train)

# Prediction
y_pred_dt = dt.predict(X_test)

# Accuracy
print("Decision Tree Accuracy:", accuracy_score(y_test, y_pred_dt))

```

Decision Tree Accuracy: 1.0

```

# Recurrent Neural Network (RNN)
import numpy as np
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import SimpleRNN, Dense

# Reshape for RNN: (samples, timesteps, features)
X_train_rnn = X_train.reshape((X_train.shape[0], 1, X_train.shape[1]))
X_test_rnn = X_test.reshape((X_test.shape[0], 1, X_test.shape[1]))

# Model
model = Sequential()
model.add(SimpleRNN(16, input_shape=(1, 4)))
model.add(Dense(3, activation='softmax'))

model.compile(optimizer='adam', loss='sparse_categorical_crossentropy', metrics=['accuracy'])

# Train
model.fit(X_train_rnn, y_train, epochs=50, batch_size=8, verbose=0)

# Evaluate
loss, acc = model.evaluate(X_test_rnn, y_test)
print("RNN Accuracy:", acc)

```

/usr/local/lib/python3.12/dist-packages/keras/src/layers/rnn/rnn.py:199: UserWarning: Do not pass an `input\_shape`/`input\_dim` a
super().__init__(**kwargs)

1/1  1s 538ms/step - accuracy: 0.9667 - loss: 0.2015
RNN Accuracy: 0.9666666388511658